



Python While loop (*condition statements*)

Syntax

- **while** \<condition>:

```
In [ ]: a = 1    # Initilization/Start
        while a<=10:    # Condition/End
            print(a,end=" ")
            a += 1        #Incrementation/Jump
        print("\nLoop Over")
```

```
In [ ]: a = 0
        while a<=100:
            print(a,end=" ")
            a += 10

        print("\nLoop Over")
```

```
In [ ]: # Mohamed FaizalJust now

        a = 100
        while a >= 0:
            print(a, end = " ")
            a -= 10
```

```
In [ ]: a = 1
        while a<=10:
            print(a,end=" ")
            a += 1

        print("\nLoop Over")
```

break - stop the loop

```
In [ ]: # Infinite Loop
        a = 1

        while a>0:
            print(a,end=" ")
            a += 1
```

```
In [ ]: a = 1

        while a>0:
            if a == 5:
                print(a)
                break
            print(a,end=" ")
```

```
a += 1
```

continue - go back to the top of my loop

```
In [ ]: a = 1
while a<=20:
    if a % 3 == 0:
        print(a,end=" ")
        a += 1
        continue

    a += 1

print("\nLoop over")
```

Nested while Loop

```
In [9]: """
12345
12345
12345
12345
12345
"""

a = 1
while a<=5:
    b = 1
    while b<=5:
        print(b,end=" ")
        b += 1
    a += 1
    print()
```

```
12345
12345
12345
12345
12345
```

```
In [14... a = 1
while a<=3:
    b = 1
    while b<=3:
        print(b,end=" ")
        b += 1
    a += 1
    print()
```

```
123
123
123
```

```
In [ ]: # Task
row = int(input("Enter a roe Number"))
column = int(input("Enter a coumn Number"))
```

```
In [5]: # task

# Venkatesh Balasubramanian

row = int(input("Enter a roe Number"))
column = int(input("Enter a coumn Number"))

a = 1
while a<=row:
    b = 1
    while b<=column:
        print(b,end="")
        b +=1
    a +=1
    print()
    # by venkat
```

```
12345678
12345678
12345678
12345678
12345678
```

```
In [ ]: # Baby GaneshJust now
# task
row=int(input("enter a row number"))
cloumn=int(input("enter a cloumn number"))
i = 1
while i <= row:
    j = 1
    while j <= cloumn:
        print(j, end=" ")
        j += 1
    print()
    i += 1
```

```
In [ ]: # Mohamed Faiza
row = int(input("enter number of rows: "))
column = int(input("enter number of columns: "))
a = 1
while a <= row:
    b = 1
    while b <= column:
        print(b, end = "")
        b += 1
    a += 1
    print()
```

```
In [7]: # David Raj
```

```

row = int(input("Enter a row number: "))
column = int(input("Enter a column number: "))
i = 1
while row>=i:
    j = 1
    while column>=j:
        print(j,end=" ")
        j += 1
    i += 1
    print()

```

```

123456
123456
123456
123456
123456

```

In [9]: *# chakravarthy K*

```

a=int(input())
b=int(input())

x =1
while x<=a:
    y=1
    while y<=b:
        print(y,end=" ")
        y+=1
    x+=1
    print()

```

```

12345678910
12345678910
12345678910
12345678910
12345678910
12345678910
12345678910
12345678910
12345678910
12345678910

```

Python for Loop

Syntax

for \<variable_name> in \<str/list/tuple/set/dict>:

```

In [14... mystr = "Hello world"

for x in mystr:    # Fetches Letter by letter
    print(x,end=" ")

```

Hello world

```
In [18... mylist = [10,20.55,"hello",True,20.55,"hello"]
```

```
for x in mylist:  
    print(x,end=",")
```

10,20.55,hello,True,20.55,hello,

range() - is a python standard function used for numbers

```
In [37... for x in range(10+1): # end before  
    print(x,end=' ')
```

0 1 2 3 4 5 6 7 8 9 10

```
In [38... for x in range(5,10+1): # start with end before  
    print(x,end=' ')
```

5 6 7 8 9 10

```
In [ ]: num = int(input("enter a number :"))
```

```
for x in range(0,num+1):  
    print(x,end=' ')
```

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

```
In [47... for x in range(100,1000+1,100): # Jump  
    print(x,end=" ")
```

100 200 300 400 500 600 700 800 900 1000

```
In [ ]: for x in range(100,0,-10): # reverse  
    print(x,end=" ")
```

100 90 80 70 60 50 40 30 20 10

for with else

```
In [52... for x in "hello world":  
    print(x,end='')
```

hello world

```
In [53... # If Loop completes successfully, else will worke  
# If loop is unsuccessfully , else will not worke
```

```
for x in "Hello world":  
    print(x,end = "")  
else:  
    print("\nThe above loop complieted successfully")
```

Hello world

The above loop complieted successfully

```
In [55... for x in "Helloworld":
```

```

    if x == ' ':
        break
    print(x,end = "")
else:
    print("\nThe above loop completed successfully")

```

Hello world

The above loop completed successfully

In [60... *# find if a number is divisible by 5 in the list*

```

mylist = [2,6,9,10,15,11,16,97]

for each in mylist:
    if each % 5 == 0:
        print(each,"Divisible By 5")
        break
else:
    print("Number divisible by 5 not found")

```

10 Divisible By 5

Nested Loop

In [64...

```

"""
12345
12345
12345
12345
12345
"""

for x in range(5): # Row
    for y in range(1,5+1):
        print(y,end='')
    print()

```

12345
12345
12345
12345
12345

In [2]:

```

for x in ["Yellow","Green","Red"]:

    print("-----")
    for y in ["Cat","Dog","Car","House"]:
        print(x,y)
    print("-----")

print("Loop Over")

```

Yellow Cat
Yellow Dog
Yellow Car
Yellow House

Green Cat
Green Dog
Green Car
Green House

Red Cat
Red Dog
Red Car
Red House

Loop Over